

Exercise 1: Implementing the Singleton Pattern

CODE:

Singleton Class:

```
Module 1 > SingletonPatternDemo.java
1
2 public class SingletonPatternDemo {
3     public static void main(String[] args) {
4         System.out.println("=== Testing Singleton Pattern Implementation ===");
5
6         // 1. Attempt to get the instance of Logger for the first time
7         Logger logger1 = Logger.getInstance();
8         logger1.log("First log message using logger1.");
9
10        // 2. Attempt to get the instance of Logger for the second time
11        Logger logger2 = Logger.getInstance();
12        logger2.log("Second log message using logger2.");
13
14        // 3. Verify that both references point to the exact same instance
15        System.out.println("\n--- Verification ---");
16        System.out.println("Logger 1 Hashcode: " + System.identityHashCode(logger1));
17        System.out.println("Logger 2 Hashcode: " + System.identityHashCode(logger2));
18
19        boolean areIdentical = (logger1 == logger2);
20        System.out.println("Are logger1 and logger2 references pointing to the same instance? " + areIdentical);
21
22        if (areIdentical) {
23            System.out.println("SUCCESS: Singleton pattern is working correctly! Only one instance exists.");
24        } else {
25            System.out.println("FAILURE: Singleton pattern failed. Multiple instances found.");
26        }
27    }
28 }
29
```

Logger Class:

Module 1 > Logger.java

```
1
2 public class Logger {
3     private static volatile Logger instance;
4
5     private Logger() {
6         if (instance != null) {
7             throw new IllegalStateException("Instance already created. Use getInstance()
8             method.");
9         }
10    }
11
12    public static Logger getInstance() {
13        if (instance == null) {
14            synchronized (Logger.class) {
15                if (instance == null) {
16                    instance = new Logger();
17                }
18            }
19            return instance;
20        }
21
22        public void log(String message) {
23            System.out.println("[LOG] " + java.time.LocalDateTime.now() + " - " + message);
24        }
25    }
26 }
```

OUTPUT:

Problems Output Debug Console Terminal Ports

- PS D:\Github Projects\Cognizant-Hands-On-Exercise> cd "d:\Github Projects\Cognizant-Hands-On-Exercise\Module 1\" ; if (\$?) { javac SingletonPatternDemo.java } ; if (\$?) { java SingletonPatternDemo }
=== Testing Singleton Pattern Implementation ===
[LOG] 2026-06-20T21:19:15.517670200 - First log message using logger1.
[LOG] 2026-06-20T21:19:15.520662900 - Second log message using logger2.

--- Verification ---
Logger 1 Hashcode: 2018699554
Logger 2 Hashcode: 2018699554
Are logger1 and logger2 references pointing to the same instance? true
SUCCESS: Singleton pattern is working correctly! Only one instance exists.
- PS D:\Github Projects\Cognizant-Hands-On-Exercise\Module 1>